

Phase 3 & Phase 4 Completion Update

Sports Club Management System — CS323 Database Systems

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We have completed Phase 3, our logical design and normalization work: we resolved all eight open design decisions, brought the schema to Third Normal Form, and corrected one delete-rule inconsistency we caught while cross-checking our own documentation. In Phase 4 we have written all thirteen tables and our database-creation script; indexes, triggers, and privilege grants are still ahead of us.

1. Phase 3 — Logical Design & Normalization — complete

We converted the Phase 2 ER model into a relational schema and normalized it to 3NF. We made six changes along the way: we split names and contact info into atomic attributes (1NF, C1); we extracted `MEMBERSHIP_TYPE` to remove a transitive dependency on fee (3NF, C2); we added a price-snapshot column to protect historical revenue from later price changes (C3); we resolved our two M:N relationships into junction tables (C4); we added a `SPORT` lookup table for extensibility (C5); and we added `APP_USER` so role-based access and password hashing have somewhere to live (C6).

Design decisions (D1–D8) — we resolved all eight

#	Decision	Our resolution
D1	Add <code>APP_USER</code> ?	Yes — FR10/NFR2 require it.
D2	<code>TEAM.sport</code>	<code>SPORT</code> lookup table — NFR5 (grow without redesign).
D3	<code>FACILITY_BOOKING</code> time model	Fixed 2-hour slots, 06:00–22:00; BR5 enforced by <code>UNIQUE(facility_id, booking_date, time_slot)</code> .
D4	Junction table keys	Composite PKs — self-enforcing against duplicate rows.
D5	Delete rules	RESTRICT on financial/historical links; CASCADE only on our two weak entities.
D6	<code>PAYMENT.payment_date</code>	DATETIME — recovering a lost time isn't possible from DATE.
D7	<code>COMPETITION.venue</code>	Free-text <code>VARCHAR(100)</code> , nullable (venue can be TBD).
D8	Key style	Surrogate <code>INT AUTO_INCREMENT</code> throughout.

A correction we made during cross-check

While reviewing our design document against our own ER diagram, we caught an inconsistency in our delete rules. We had initially set `FACILITY_BOOKING → TEAM` to CASCADE, but our own Phase 3 report reserves CASCADE for our two *weak* entities — `TEAM_ROSTER` and `TEAM_COMPETITION` — since they have no identity independent of their parents. `FACILITY_BOOKING` has its own surrogate key, making it a strong entity by our own test, so we corrected that foreign key to RESTRICT: a disbanded team can no longer silently erase its booking history, and must have its bookings explicitly reassigned or removed first.

2. Phase 4 — DDL — tables complete, indexes pending

We wrote and reviewed all thirteen tables individually, in dependency order, then committed them to `schema/02_tables.sql`. We also wrote `schema/01_create_database.sql`, a re-runnable drop-and-recreate script for our `sports_club` database. Our schema now stands at **13 tables, 12 foreign keys, and 8 named CHECK constraints**, with every table declaring `ENGINE=InnoDB` and `utf8mb4/utf8mb4_unicode_ci`. We verified this against our actual MariaDB build (10.4.28 — comfortably past the 10.2.1 threshold that both CHECK enforcement and `DEFAULT CURRENT_DATE` require).

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|-----------------------------------|-----------------------------------|--------------------------|
| 1. <code>membership_type</code> | 12. <code>team_roster</code> | 9. <code>app_user</code> |
| 6. <code>competition</code> | 3. <code>athlete</code> | 5. <code>facility</code> |
| 11. <code>facility_booking</code> | 8. <code>team</code> | 10. <code>payment</code> |
| 2. <code>sport</code> | 13. <code>team_competition</code> | |
| 7. <code>membership</code> | 4. <code>coach</code> | |

Where our business rules stand

Rule	Mechanism	Status
BR1 — membership before team join	BEFORE INSERT trigger	Phase 6b
BR2 — membership status is one of three values	ENUM + NOT NULL	Done
BR3 — payment linked to exactly one membership	NOT NULL FK	Done
BR4 — team has exactly one coach	NOT NULL FK	Done
BR5 — no facility double-booking	UNIQUE(facility_id, booking_date, time_slot)	Done — no trigger needed
BR6 — teams ↔ competitions, M:N	team_competition junction	Done
BR7 — only staff/admin change payment/status	app_user.role + GRANT	Partial — GRANT is Phase 7

3. Repository & what's left

We pushed our work to branch `feature/phase4-ddl` (commit: Database and Table Creation) and opened a pull request against `main` — we don't push directly to `main`, per our own team convention. We updated `docs/Phase3_Logical_Design.md` with all eight resolutions and our delete-rule correction.

What's left, across both phases:

- `schema/03_indexes.sql` — our eight planned indexes, not yet written
- Live execution and verification against a running MariaDB instance — our scripts are unexecuted pending an XAMPP service issue; our verification standard requires proving constraints reject bad data, not just a clean run
- Phase 6b: the BR1 trigger and our single-active-membership rule on `MEMBERSHIP`
- Phase 7: GRANT privileges completing BR7
- Redraw our ER diagram to reflect our current 13-table schema
- Tag `v1-schema-locked` once merged